

Microsoft Windows [Version 10.0.26200.9168]
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```
C:\Users\LENOVO>mongosh "mongodb+srv://cluster0.yzn22yh.mongodb.net/" --
apiVersion 1 --username jsaishashank_db_user
Enter password: *****
Current Mongosh Log ID: 6a8bfc53d7120b9157161b7d
Connecting to:
mongodb+srv://<credentials>@cluster0.yzn22yh.mongodb.net/?appName=mongosh
+2.10.0
Using MongoDB:      8.0.29 (API Version 1)
Using Mongosh:      2.10.0
```

For mongosh info see: <https://www.mongodb.com/docs/mongodb-shell/>

```
Atlas atlas-ios9hn-shard-0 [primary] test> use BookFlow
switched to db BookFlow
Atlas atlas-ios9hn-shard-0 [primary] BookFlow> switched to db BookFlow
Uncaught:
SyntaxError: Missing semicolon. (1:8)
```

```
> 1 | switched to db BookFlow
    |           ^
    2 |
```

```
Atlas atlas-ios9hn-shard-0 [primary] BookFlow>
db.createCollection("book_metadata")
{ ok: 1 }
Atlas atlas-ios9hn-shard-0 [primary] BookFlow>
db.book_metadata.insertOne({
  |   title: "The Silent Algorithm",
  |   author: "R. Menon",
  |   genre: "Technology",
  |   publishedYear: 2022,
  |   formats: ["Hardcover", "Digital", "Audiobook"],
  |   specs: {
  |     pages: 340,
  |     fileSize_MB: 4.2,
  |     drm: true,
  |     narrator: "K. Rao",
  |     runtime_min: 610
  |   },
  |   avgRating: 4.6,
  |   reviews: [
  |     {
  |       member_id: "M1001",
  |       rating: 5,
  |       comment: "Excellent deep dive into digital systems design.",
  |       date: ISODate("2023-01-15")
  |     },
  |     {
  |       member_id: "M1042",
  |       rating: 4,
  |       comment: "Good but the Digital edition had formatting issues.",
```

```

|         date: ISODate("2023-02-20")
|     }
| ]
| })
| {
|     acknowledged: true,
|     insertedId: ObjectId('6a8bfceed7120b9157161b7e')
| }
Atlas atlas-ios9hn-shard-0 [primary] BookFlow>
db.book_metadata.insertMany([
|     {
|         title: "Ocean of Code",
|         author: "A. Fernandes",
|         genre: "Technology",
|         publishedYear: 2021,
|         formats: ["Digital", "Paperback"],
|         specs: {
|             pages: 210,
|             fileSize_MB: 2.1,
|             drm: false
|         },
|         avgRating: 4.2,
|         reviews: [
|             {
|                 member_id: "M2001",
|                 rating: 4,
|                 comment: "Solid Digital read, clean layout.",
|                 date: ISODate("2022-05-10")
|             },
|             {
|                 member_id: "M2002",
|                 rating: 5,
|                 comment: "Best programming book this year.",
|                 date: ISODate("2022-06-01")
|             }
|         ]
|     },
|     {
|         title: "Whispers of Yesterday",
|         author: "S. Kapoor",
|         genre: "Fiction",
|         publishedYear: 2019,
|         formats: ["Hardcover", "Paperback"],
|         specs: {
|             pages: 410,
|             binding: "Sewn"
|         },
|         avgRating: 3.8,
|         reviews: [
|             {
|                 member_id: "M3001",
|                 rating: 3,
|                 comment: "Slow start but strong ending.",

```

```

|         date: ISODate("2020-01-15")
|     },
|     {
|         member_id: "M3002",
|         rating: 4,
|         comment: "Beautifully written prose.",
|         date: ISODate("2020-02-11")
|     }
| ]
| },
|
| {
|     title: "Quantum Horizons",
|     author: "R. Menon",
|     genre: "Science",
|     publishedYear: 2023,
|     formats: ["Digital", "Audiobook"],
|     specs: {
|         fileSize_MB: 5.6,
|         drm: true,
|         narrator: "P. Iyer",
|         runtime_min: 540
|     },
|     avgRating: 4.9,
|     reviews: [
|         {
|             member_id: "M4001",
|             rating: 5,
|             comment: "The Digital format made annotations so easy.",
|             date: ISODate("2023-08-01")
|         },
|         {
|             member_id: "M4002",
|             rating: 5,
|             comment: "Mind-bending and well narrated.",
|             date: ISODate("2023-09-14")
|         }
|     ]
| }
| })
| {
|     acknowledged: true,
|     insertedIds: {
|         '0': ObjectId('6a8bfcf8d7120b9157161b7f'),
|         '1': ObjectId('6a8bfcf8d7120b9157161b80'),
|         '2': ObjectId('6a8bfcf8d7120b9157161b81')
|     }
| }
Atlas atlas-ios9hn-shard-0 [primary] BookFlow>
db.book_metadata.find().pretty()
[
  {
    _id: ObjectId('6a8bfceed7120b9157161b7e'),
    title: 'The Silent Algorithm',

```

```

author: 'R. Menon',
genre: 'Technology',
publishedYear: 2022,
formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
specs: {
  pages: 340,
  fileSize_MB: 4.2,
  drm: true,
  narrator: 'K. Rao',
  runtime_min: 610
},
avgRating: 4.6,
reviews: [
  {
    member_id: 'M1001',
    rating: 5,
    comment: 'Excellent deep dive into digital systems design.',
    date: ISODate('2023-01-15T00:00:00.000Z')
  },
  {
    member_id: 'M1042',
    rating: 4,
    comment: 'Good but the Digital edition had formatting issues.',
    date: ISODate('2023-02-20T00:00:00.000Z')
  }
]
},
{
  _id: ObjectId('6a8bfcf8d7120b9157161b7f'),
  title: 'Ocean of Code',
  author: 'A. Fernandes',
  genre: 'Technology',
  publishedYear: 2021,
  formats: [ 'Digital', 'Paperback' ],
  specs: { pages: 210, fileSize_MB: 2.1, drm: false },
  avgRating: 4.2,
  reviews: [
    {
      member_id: 'M2001',
      rating: 4,
      comment: 'Solid Digital read, clean layout.',
      date: ISODate('2022-05-10T00:00:00.000Z')
    },
    {
      member_id: 'M2002',
      rating: 5,
      comment: 'Best programming book this year.',
      date: ISODate('2022-06-01T00:00:00.000Z')
    }
  ]
},
{
  _id: ObjectId('6a8bfcf8d7120b9157161b80'),
  title: 'Whispers of Yesterday',

```

```

    author: 'S. Kapoor',
    genre: 'Fiction',
    publishedYear: 2019,
    formats: [ 'Hardcover', 'Paperback' ],
    specs: { pages: 410, binding: 'Sewn' },
    avgRating: 3.8,
    reviews: [
      {
        member_id: 'M3001',
        rating: 3,
        comment: 'Slow start but strong ending.',
        date: ISODate('2020-01-15T00:00:00.000Z')
      },
      {
        member_id: 'M3002',
        rating: 4,
        comment: 'Beautifully written prose.',
        date: ISODate('2020-02-11T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a8bfcf8d7120b9157161b81'),
    title: 'Quantum Horizons',
    author: 'R. Menon',
    genre: 'Science',
    publishedYear: 2023,
    formats: [ 'Digital', 'Audiobook' ],
    specs: { fileSize_MB: 5.6, drm: true, narrator: 'P. Iyer',
runtime_min: 540 },
    avgRating: 4.9,
    reviews: [
      {
        member_id: 'M4001',
        rating: 5,
        comment: 'The Digital format made annotations so easy.',
        date: ISODate('2023-08-01T00:00:00.000Z')
      },
      {
        member_id: 'M4002',
        rating: 5,
        comment: 'Mind-bending and well narrated.',
        date: ISODate('2023-09-14T00:00:00.000Z')
      }
    ]
  }
]
Atlas atlas-ios9hn-shard-0 [primary] BookFlow> db.book_metadata.find(
|   { avgRating: { $gt: 4 } },
|   { title: 1, avgRating: 1, _id: 0 }
| )
[
  { title: 'The Silent Algorithm', avgRating: 4.6 },
  { title: 'Ocean of Code', avgRating: 4.2 },

```

```

    { title: 'Quantum Horizons', avgRating: 4.9 }
  ]
Atlas atlas-ios9hn-shard-0 [primary] BookFlow> The Silent Algorithm
4.6
| Ocean of Code          4.2
| Quantum Horizons      4.9
Uncaught:
SyntaxError: Missing semicolon. (1:3)

```

```

> 1 | The Silent Algorithm      4.6
    |      ^
    2 | Ocean of Code          4.2
    3 | Quantum Horizons      4.9
    4 |

```

```

Atlas atlas-ios9hn-shard-0 [primary] BookFlow> db.book_metadata.find(
|   { formats: { $in: ["Digital", "Audiobook"] } },
|   { title: 1, formats: 1, _id: 0 }
| )
[
  {
    title: 'The Silent Algorithm',
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ]
  },
  { title: 'Ocean of Code', formats: [ 'Digital', 'Paperback' ] },
  { title: 'Quantum Horizons', formats: [ 'Digital', 'Audiobook' ] }
]

```

```

Atlas atlas-ios9hn-shard-0 [primary] BookFlow> db.book_metadata.find({
|   avgRating: { $gt: 4 },
|   formats: { $in: ["Digital"] }
| })
[
  {
    _id: ObjectId('6a8bfceed7120b9157161b7e'),
    title: 'The Silent Algorithm',
    author: 'R. Menon',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      },
    ],
  },
]

```

```

    {
      member_id: 'M1042',
      rating: 4,
      comment: 'Good but the Digital edition had formatting issues.',
      date: ISODate('2023-02-20T00:00:00.000Z')
    }
  ]
},
{
  _id: ObjectId('6a8bfcf8d7120b9157161b7f'),
  title: 'Ocean of Code',
  author: 'A. Fernandes',
  genre: 'Technology',
  publishedYear: 2021,
  formats: [ 'Digital', 'Paperback' ],
  specs: { pages: 210, fileSize_MB: 2.1, drm: false },
  avgRating: 4.2,
  reviews: [
    {
      member_id: 'M2001',
      rating: 4,
      comment: 'Solid Digital read, clean layout.',
      date: ISODate('2022-05-10T00:00:00.000Z')
    },
    {
      member_id: 'M2002',
      rating: 5,
      comment: 'Best programming book this year.',
      date: ISODate('2022-06-01T00:00:00.000Z')
    }
  ]
},
{
  _id: ObjectId('6a8bfcf8d7120b9157161b81'),
  title: 'Quantum Horizons',
  author: 'R. Menon',
  genre: 'Science',
  publishedYear: 2023,
  formats: [ 'Digital', 'Audiobook' ],
  specs: { fileSize_MB: 5.6, drm: true, narrator: 'P. Iyer',
runtime_min: 540 },
  avgRating: 4.9,
  reviews: [
    {
      member_id: 'M4001',
      rating: 5,
      comment: 'The Digital format made annotations so easy.',
      date: ISODate('2023-08-01T00:00:00.000Z')
    },
    {
      member_id: 'M4002',
      rating: 5,
      comment: 'Mind-bending and well narrated.',
      date: ISODate('2023-09-14T00:00:00.000Z')
    }
  ]
}

```

```

    }
  ]
}
]
Atlas atlas-ios9hn-shard-0 [primary] BookFlow> db.book_metadata.Atlas
atlas-ios9hn-shard-0 [primary] BookFlow> db.book_metadata.find(
|   { "reviews.comment": { $regex: "Digital", $options: "i" } },
|   { title: 1, "reviews.$": 1 }
| )
[
  {
    _id: ObjectId('6a8bfceed7120b9157161b7e'),
    title: 'The Silent Algorithm',
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a8bfcf8d7120b9157161b7f'),
    title: 'Ocean of Code',
    reviews: [
      {
        member_id: 'M2001',
        rating: 4,
        comment: 'Solid Digital read, clean layout.',
        date: ISODate('2022-05-10T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a8bfcf8d7120b9157161b81'),
    title: 'Quantum Horizons',
    reviews: [
      {
        member_id: 'M4001',
        rating: 5,
        comment: 'The Digital format made annotations so easy.',
        date: ISODate('2023-08-01T00:00:00.000Z')
      }
    ]
  }
]
Atlas atlas-ios9hn-shard-0 [primary] BookFlow>
db.book_metadata.aggregate([
|   {
|     $match: {
|       publishedYear: { $gt: 2020 }
|     }
|   },
| ],

```



```

|
|   {
|     $unwind: "$reviews"
|   },
|
|   {
|     $group: {
|       _id: "$title",
|       avgReviewRating: { $avg: "$reviews.rating" },
|       totalReviews: { $sum: 1 },
|       author: { $first: "$author" }
|     }
|   },
|
|   {
|     $sort: {
|       avgReviewRating: -1
|     }
|   },
|
|   {
|     $project: {
|       _id: 0,
|       title: "$_id",
|       author: 1,
|       avgReviewRating: {
|         $round: ["$avgReviewRating", 2]
|       },
|       totalReviews: 1
|     }
|   }
| ] )
[
  {
    totalReviews: 2,
    author: 'R. Menon',
    title: 'Quantum Horizons',
    avgReviewRating: 5
  },
  {
    totalReviews: 2,
    author: 'R. Menon',
    title: 'The Silent Algorithm',
    avgReviewRating: 4.5
  },
  {
    totalReviews: 2,
    author: 'A. Fernandes',
    title: 'Ocean of Code',
    avgReviewRating: 4.5
  }
]
Atlas atlas-ios9hn-shard-0 [primary] BookFlow> $match â†’ $unwind â†’
$group â†’ $sort â†’ $project

```

Uncaught:

SyntaxError: Unexpected character 'â+'. (1:7)

```
> 1 | $match â+ $unwind â+ $group â+ $sort â+ $project
    |           ^
    2 |
```

Atlas atlas-ios9hn-shard-0 [primary] BookFlow> Quantum Horizons

| Average Rating: 5

| Total Reviews: 2

|

| Ocean of Code

| Average Rating: 4.5

| Total Reviews: 2

Uncaught:

SyntaxError: Missing semicolon. (1:7)

```
> 1 | Quantum Horizons
    |           ^
    2 | Average Rating: 5
    3 | Total Reviews: 2
    4 |
```

Atlas atlas-ios9hn-shard-0 [primary] BookFlow>

db.book_metadata.createIndex({

| "reviews.comment": "text"

| })

reviews.comment_text

Atlas atlas-ios9hn-shard-0 [primary] BookFlow> db.book_metadata.find({

| \$text: {

| \$search: "Digital"

| }

| })

[

{

_id: ObjectId('6a8bfceed7120b9157161b7e'),

title: 'The Silent Algorithm',

author: 'R. Menon',

genre: 'Technology',

publishedYear: 2022,

formats: ['Hardcover', 'Digital', 'Audiobook'],

specs: {

pages: 340,

fileSize_MB: 4.2,

drm: true,

narrator: 'K. Rao',

runtime_min: 610

},

avgRating: 4.6,

reviews: [

{

member_id: 'M1001',

rating: 5,

comment: 'Excellent deep dive into digital systems design.',

```

        date: ISODate('2023-01-15T00:00:00.000Z')
    },
    {
        member_id: 'M1042',
        rating: 4,
        comment: 'Good but the Digital edition had formatting issues.',
        date: ISODate('2023-02-20T00:00:00.000Z')
    }
]
},
{
    _id: ObjectId('6a8bfcf8d7120b9157161b81'),
    title: 'Quantum Horizons',
    author: 'R. Menon',
    genre: 'Science',
    publishedYear: 2023,
    formats: [ 'Digital', 'Audiobook' ],
    specs: { fileSize_MB: 5.6, drm: true, narrator: 'P. Iyer',
runtime_min: 540 },
    avgRating: 4.9,
    reviews: [
        {
            member_id: 'M4001',
            rating: 5,
            comment: 'The Digital format made annotations so easy.',
            date: ISODate('2023-08-01T00:00:00.000Z')
        },
        {
            member_id: 'M4002',
            rating: 5,
            comment: 'Mind-bending and well narrated.',
            date: ISODate('2023-09-14T00:00:00.000Z')
        }
    ]
},
{
    _id: ObjectId('6a8bfcf8d7120b9157161b7f'),
    title: 'Ocean of Code',
    author: 'A. Fernandes',
    genre: 'Technology',
    publishedYear: 2021,
    formats: [ 'Digital', 'Paperback' ],
    specs: { pages: 210, fileSize_MB: 2.1, drm: false },
    avgRating: 4.2,
    reviews: [
        {
            member_id: 'M2001',
            rating: 4,
            comment: 'Solid Digital read, clean layout.',
            date: ISODate('2022-05-10T00:00:00.000Z')
        },
        {
            member_id: 'M2002',
            rating: 5,

```

```

        comment: 'Best programming book this year.',
        date: ISODate('2022-06-01T00:00:00.000Z')
    }
]
}
]
Atlas atlas-ios9hn-shard-0 [primary] BookFlow> db.book_metadata.find(
|   {
|     $text: {
|       $search: "Digital"
|     }
|   },
|   {
|     title: 1,
|     score: {
|       $meta: "textScore"
|     }
|   }
| ).sort({
|   score: {
|     $meta: "textScore"
|   }
| })
[
  {
    _id: ObjectId('6a8bfceed7120b9157161b7e'),
    title: 'The Silent Algorithm',
    score: 1.1833333333333333
  },
  {
    _id: ObjectId('6a8bfcf8d7120b9157161b7f'),
    title: 'Ocean of Code',
    score: 0.6
  },
  {
    _id: ObjectId('6a8bfcf8d7120b9157161b81'),
    title: 'Quantum Horizons',
    score: 0.6
  }
]
Atlas atlas-ios9hn-shard-0 [primary] BookFlow> db.book_metadata.find({
|   $text: {
|     $search: "Digital"
|   }
| }).explain("executionStats")
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'BookFlow.book_metadata',
    parsedQuery: {
      '$text': {
        '$search': 'Digital',
        '$language': 'english',
        '$caseSensitive': false,

```

```

        '$diacriticSensitive': false
    }
},
indexFilterSet: false,
queryHash: 'CF6F4CEE',
planCacheShapeHash: 'CF6F4CEE',
planCacheKey: '08852285',
optimizationTimeMillis: 0,
maxIndexedOrSolutionsReached: false,
maxIndexedAndSolutionsReached: false,
maxScansToExplodeReached: false,
prunedSimilarIndexes: false,
winningPlan: {
    isCached: false,
    stage: 'TEXT_MATCH',
    indexPrefix: {},
    indexName: 'reviews.comment_text',
    parsedTextQuery: {
        terms: [ 'digit' ],
        negatedTerms: [],
        phrases: [],
        negatedPhrases: []
    },
    textIndexVersion: 3,
    inputStage: {
        stage: 'FETCH',
        inputStage: {
            stage: 'IXSCAN',
            keyPattern: { _fts: 'text', _ftsx: 1 },
            indexName: 'reviews.comment_text',
            isMultiKey: true,
            isUnique: false,
            isSparse: false,
            isPartial: false,
            indexVersion: 2,
            direction: 'backward',
            indexBounds: {}
        }
    }
},
rejectedPlans: []
},
executionStats: {
    executionSuccess: true,
    nReturned: 3,
    executionTimeMillis: 0,
    totalKeysExamined: 3,
    totalDocsExamined: 3,
    executionStages: {
        isCached: false,
        stage: 'TEXT_MATCH',
        nReturned: 3,
        executionTimeMillisEstimate: 0,
        works: 4,

```

```
advanced: 3,
needTime: 0,
needYield: 0,
saveState: 0,
restoreState: 0,
isEOF: 1,
indexPrefix: {},
indexName: 'reviews.comment_text',
parsedTextQuery: {
  terms: [ 'digit' ],
  negatedTerms: [],
  phrases: [],
  negatedPhrases: []
},
textIndexVersion: 3,
docsRejected: 0,
inputStage: {
  stage: 'FETCH',
  nReturned: 3,
  executionTimeMillisEstimate: 0,
  works: 4,
  advanced: 3,
  needTime: 0,
  needYield: 0,
  saveState: 0,
  restoreState: 0,
  isEOF: 1,
  docsExamined: 3,
  alreadyHasObj: 0,
  inputStage: {
    stage: 'IXSCAN',
    nReturned: 3,
    executionTimeMillisEstimate: 0,
    works: 4,
    advanced: 3,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    keyPattern: { _fts: 'text', _ftsx: 1 },
    indexName: 'reviews.comment_text',
    isMultiKey: true,
    isUnique: false,
    isSparse: false,
    isPartial: false,
    indexVersion: 2,
    direction: 'backward',
    indexBounds: {},
    keysExamined: 3,
    seeks: 1,
    dupsTested: 3,
    dupsDropped: 0
  }
}
```

```

    }
  }
},
queryShapeHash:
'1D1348BE410691F8B41571992B30363DC0E9797E4ED907646E00B162CCEFF38E',
command: {
  find: 'book_metadata',
  filter: { '$text': { '$search': 'Digital' } },
  '$db': 'BookFlow'
},
serverInfo: {
  host: 'ac-nrqixna-shard-00-01.yzn22yh.mongodb.net',
  port: 27017,
  version: '8.0.29',
  gitVersion: '559d67c651f7393e62757234a0b25cb7a8622148'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 16793600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 33554432,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1,
'$clusterTime': {
  clusterTime: Timestamp({ t: 1787559260, i: 20 }),
  signature: {
    hash: Binary.createFromBase64('Z8VMbO3ZbvU4zUkzRThc3mGWY2M=', 0),
    keyId: Long('7617890356549386242')
  }
},
operationTime: Timestamp({ t: 1787559260, i: 20 })
}
Atlas atlas-ios9hn-shard-0 [primary] BookFlow> db.book_metadata.find({
|   $text: {
|     $search: "Digital"
|   }
| }).explain("executionStats")
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'BookFlow.book_metadata',
    parsedQuery: {
      '$text': {
        '$search': 'Digital',
        '$language': 'english',
        '$caseSensitive': false,
        '$diacriticSensitive': false
      }
    }
  }
}

```

```

    },
    indexFilterSet: false,
    queryHash: 'CF6F4CEE',
    planCacheShapeHash: 'CF6F4CEE',
    planCacheKey: '08852285',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'TEXT_MATCH',
      indexPrefix: {},
      indexName: 'reviews.comment_text',
      parsedTextQuery: {
        terms: [ 'digit' ],
        negatedTerms: [],
        phrases: [],
        negatedPhrases: []
      },
    },
    textIndexVersion: 3,
    inputStage: {
      stage: 'FETCH',
      inputStage: {
        stage: 'IXSCAN',
        keyPattern: { _fts: 'text', _ftsx: 1 },
        indexName: 'reviews.comment_text',
        isMultiKey: true,
        isUnique: false,
        isSparse: false,
        isPartial: false,
        indexVersion: 2,
        direction: 'backward',
        indexBounds: {}
      }
    },
    },
    rejectedPlans: []
  },
  executionStats: {
    executionSuccess: true,
    nReturned: 3,
    executionTimeMillis: 0,
    totalKeysExamined: 3,
    totalDocsExamined: 3,
    executionStages: {
      isCached: false,
      stage: 'TEXT_MATCH',
      nReturned: 3,
      executionTimeMillisEstimate: 0,
      works: 4,
      advanced: 3,
      needTime: 0,

```



```

needYield: 0,
saveState: 0,
restoreState: 0,
isEOF: 1,
indexPrefix: {},
indexName: 'reviews.comment_text',
parsedTextQuery: {
  terms: [ 'digit' ],
  negatedTerms: [],
  phrases: [],
  negatedPhrases: []
},
textIndexVersion: 3,
docsRejected: 0,
inputStage: {
  stage: 'FETCH',
  nReturned: 3,
  executionTimeMillisEstimate: 0,
  works: 4,
  advanced: 3,
  needTime: 0,
  needYield: 0,
  saveState: 0,
  restoreState: 0,
  isEOF: 1,
  docsExamined: 3,
  alreadyHasObj: 0,
  inputStage: {
    stage: 'IXSCAN',
    nReturned: 3,
    executionTimeMillisEstimate: 0,
    works: 4,
    advanced: 3,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    keyPattern: { _fts: 'text', _ftsx: 1 },
    indexName: 'reviews.comment_text',
    isMultiKey: true,
    isUnique: false,
    isSparse: false,
    isPartial: false,
    indexVersion: 2,
    direction: 'backward',
    indexBounds: {},
    keysExamined: 3,
    seeks: 1,
    dupsTested: 3,
    dupsDropped: 0
  }
}
}
}

```

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    },
    queryShapeHash:
'1D1348BE410691F8B41571992B30363DC0E9797E4ED907646E00B162CCEFF38E',
    command: {
      find: 'book_metadata',
      filter: { '$text': { '$search': 'Digital' } },
      '$db': 'BookFlow'
    },
    serverInfo: {
      host: 'ac-nrqixna-shard-00-01.yzn22yh.mongodb.net',
      port: 27017,
      version: '8.0.29',
      gitVersion: '559d67c651f7393e62757234a0b25cb7a8622148'
    },
    serverParameters: {
      internalQueryFacetBufferSizeBytes: 104857600,
      internalQueryFacetMaxOutputDocSizeBytes: 104857600,
      internalLookupStageIntermediateDocumentMaxSizeBytes: 16793600,
      internalDocumentSourceGroupMaxMemoryBytes: 104857600,
      internalQueryMaxBlockingSortMemoryUsageBytes: 33554432,
      internalQueryProhibitBlockingMergeOnMongoS: 0,
      internalQueryMaxAddToSetBytes: 104857600,
      internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
      internalQueryFrameworkControl: 'trySbeRestricted',
      internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
    },
    ok: 1,
    '$clusterTime': {
      clusterTime: Timestamp({ t: 1787559282, i: 1 }),
      signature: {
        hash: Binary.createFromBase64('DC00SouulrlGIDqnuC0zUA+o1Ek=', 0),
        keyId: Long('7617890356549386242')
      }
    },
    operationTime: Timestamp({ t: 1787559282, i: 1 })
  }
Atlas atlas-ios9hn-shard-0 [primary] BookFlow> db.book_metadata.find(
|   {
|     "reviews.comment": {
|       $regex: "Digital",
|       $options: "i"
|     }
|   }
| ).explain("executionStats")
| {
|   explainVersion: '1',
|   queryPlanner: {
|     namespace: 'BookFlow.book_metadata',
|     parsedQuery: { 'reviews.comment': { '$regex': 'Digital', '$options':
'i' } },
|     indexFilterSet: false,
|     queryHash: '9DF7CB16',
|     planCacheShapeHash: '9DF7CB16',
|     planCacheKey: 'A9183C86',

```

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    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'COLLSCAN',
      filter: { 'reviews.comment': { '$regex': 'Digital', '$options':
'i' } } },
      direction: 'forward'
    },
    rejectedPlans: []
  },
  executionStats: {
    executionSuccess: true,
    nReturned: 3,
    executionTimeMillis: 0,
    totalKeysExamined: 0,
    totalDocsExamined: 4,
    executionStages: {
      isCached: false,
      stage: 'COLLSCAN',
      filter: { 'reviews.comment': { '$regex': 'Digital', '$options':
'i' } } },
      nReturned: 3,
      executionTimeMillisEstimate: 0,
      works: 5,
      advanced: 3,
      needTime: 1,
      needYield: 0,
      saveState: 0,
      restoreState: 0,
      isEOF: 1,
      direction: 'forward',
      docsExamined: 4
    }
  },
  queryShapeHash:
'5942F23D76812C6AFE0B16860E069B938E3E9F102BA6F2523A76D8516F11CC4A',
  command: {
    find: 'book_metadata',
    filter: { 'reviews.comment': { '$regex': 'Digital', '$options':
'i' } } },
    '$db': 'BookFlow'
  },
  serverInfo: {
    host: 'ac-nrqixna-shard-00-01.yzn22yh.mongodb.net',
    port: 27017,
    version: '8.0.29',
    gitVersion: '559d67c651f7393e62757234a0b25cb7a8622148'
  },
  serverParameters: {
    internalQueryFacetBufferSizeBytes: 104857600,

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        internalQueryFacetMaxOutputDocSizeBytes: 104857600,
        internalLookupStageIntermediateDocumentMaxSizeBytes: 16793600,
        internalDocumentSourceGroupMaxMemoryBytes: 104857600,
        internalQueryMaxBlockingSortMemoryUsageBytes: 33554432,
        internalQueryProhibitBlockingMergeOnMongoS: 0,
        internalQueryMaxAddToSetBytes: 104857600,
        internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
        internalQueryFrameworkControl: 'trySbeRestricted',
        internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
    },
    ok: 1,
    '$clusterTime': {
        clusterTime: Timestamp({ t: 1787559308, i: 6 }),
        signature: {
            hash: Binary.createFromBase64('ayT4fpQfV02BKZXL+4gqlXTl+w8=', 0),
            keyId: Long('7617890356549386242')
        }
    },
    operationTime: Timestamp({ t: 1787559308, i: 6 })
}
Atlas atlas-ios9hn-shard-0 [primary] BookFlow>
db.book_metadata.createIndex({
|   formats: 1
| })
formats_1
Atlas atlas-ios9hn-shard-0 [primary] BookFlow>
db.book_metadata.getIndexes()
[
  { v: 2, key: { _id: 1 }, name: '_id_' },
  {
    v: 2,
    key: { _fts: 'text', _ftsx: 1 },
    name: 'reviews.comment_text',
    weights: { 'reviews.comment': 1 },
    default_language: 'english',
    language_override: 'language',
    textIndexVersion: 3
  },
  { v: 2, key: { formats: 1 }, name: 'formats_1' }
]
Atlas atlas-ios9hn-shard-0 [primary] BookFlow>
db.book_metadata.createIndex({
|   avgRating: -1
| })
avgRating_-1
Atlas atlas-ios9hn-shard-0 [primary] BookFlow> db.book_metadata.find(
|   {
|     avgRating: {
|       $gt: 4
|     }
|   }
| ).explain("executionStats")
{
  explainVersion: '1',

```

```

queryPlanner: {
  namespace: 'BookFlow.book_metadata',
  parsedQuery: { avgRating: { '$gt': 4 } },
  indexFilterSet: false,
  queryHash: '85BA8276',
  planCacheShapeHash: '85BA8276',
  planCacheKey: '6EB4A3FD',
  optimizationTimeMillis: 0,
  maxIndexedOrSolutionsReached: false,
  maxIndexedAndSolutionsReached: false,
  maxScansToExplodeReached: false,
  prunedSimilarIndexes: false,
  winningPlan: {
    isCached: false,
    stage: 'FETCH',
    inputStage: {
      stage: 'IXSCAN',
      keyPattern: { avgRating: -1 },
      indexName: 'avgRating_-1',
      isMultiKey: false,
      multiKeyPaths: { avgRating: [] },
      isUnique: false,
      isSparse: false,
      isPartial: false,
      indexVersion: 2,
      direction: 'forward',
      indexBounds: { avgRating: [ '[inf.0, 4)' ] }
    }
  },
  rejectedPlans: []
},
executionStats: {
  executionSuccess: true,
  nReturned: 3,
  executionTimeMillis: 1,
  totalKeysExamined: 3,
  totalDocsExamined: 3,
  executionStages: {
    isCached: false,
    stage: 'FETCH',
    nReturned: 3,
    executionTimeMillisEstimate: 0,
    works: 4,
    advanced: 3,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    docsExamined: 3,
    alreadyHasObj: 0,
    inputStage: {
      stage: 'IXSCAN',
      nReturned: 3,

```

```

        executionTimeMillisEstimate: 0,
        works: 4,
        advanced: 3,
        needTime: 0,
        needYield: 0,
        saveState: 0,
        restoreState: 0,
        isEOF: 1,
        keyPattern: { avgRating: -1 },
        indexName: 'avgRating_-1',
        isMultiKey: false,
        multiKeyPaths: { avgRating: [] },
        isUnique: false,
        isSparse: false,
        isPartial: false,
        indexVersion: 2,
        direction: 'forward',
        indexBounds: { avgRating: [ '[inf.0, 4)' ] },
        keysExamined: 3,
        seeks: 1,
        dupsTested: 0,
        dupsDropped: 0
    }
}
},
queryShapeHash:
'A6411A9AD04FD3686B68F7817C38ADA41F8B73F38F98A480EF0D651344441058',
command: {
    find: 'book_metadata',
    filter: { avgRating: { '$gt': 4 } },
    '$db': 'BookFlow'
},
serverInfo: {
    host: 'ac-nrqixna-shard-00-01.yzn22yh.mongodb.net',
    port: 27017,
    version: '8.0.29',
    gitVersion: '559d67c651f7393e62757234a0b25cb7a8622148'
},
serverParameters: {
    internalQueryFacetBufferSizeBytes: 104857600,
    internalQueryFacetMaxOutputDocSizeBytes: 104857600,
    internalLookupStageIntermediateDocumentMaxSizeBytes: 16793600,
    internalDocumentSourceGroupMaxMemoryBytes: 104857600,
    internalQueryMaxBlockingSortMemoryUsageBytes: 33554432,
    internalQueryProhibitBlockingMergeOnMongoS: 0,
    internalQueryMaxAddToSetBytes: 104857600,
    internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
    internalQueryFrameworkControl: 'trySbeRestricted',
    internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1,
'$clusterTime': {
    clusterTime: Timestamp({ t: 1787559337, i: 2 }),
    signature: {

```

```
        hash: Binary.createFromBase64('UPdn+Sst8rdjCvhG+Q2AuUi9JOM=', 0),
        keyId: Long('7617890356549386242')
    },
    operationTime: Timestamp({ t: 1787559337, i: 2 })
}
Atlas atlas-ios9hn-shard-0 [primary] BookFlow>
```